



OPEN Robust fault detection and classification in power transmission lines via ensemble machine learning models

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Transmission lines are vital for delivering electricity over long distances, yet they face reliability challenges due to faults that can disrupt power supply and pose safety risks. This research introduces a novel approach for fault detection and classification by analyzing voltage and current patterns across transmission line phases. Leveraging a comprehensive dataset of diverse fault scenarios, various machine learning algorithms—including Random Forest (RF), K-Nearest Neighbors (KNN), and Long Short-Term Memory (LSTM) networks—are evaluated. An ensemble methodology, RF-LSTM Tuned KNN, is proposed to enhance detection accuracy and robustness. Results indicate that RF-LSTM Tuned KNN achieves a remarkable accuracy of 99.96% on a multi-label dataset, outperforming RF (97.50%) and KNN (96.55%). In binary classification, KNN attains the highest accuracy of 99.85%, closely followed by RF at 99.72%. This methodology provides significant advancements in fault detection capabilities, offering valuable insights for improving grid reliability and stability, and ensuring a more resilient power supply.

Keywords Transmission lines, Fault detection, Machine learning, Ensemble learning, Power stability

Transmission lines play a critical role in the transportation of electricity over long distances; however, they are susceptible to faults that can disrupt operations and cause significant economic losses¹. The reliability of these lines heavily depends on effective fault identification and classification. The power system is divided into three main stages: generation, transmission, and distribution. Each stage is designed to ensure a stable and continuous electricity supply². Due to their exposure to the environment, parts of the electrical system are more prone to faults, making fault management essential. Transmission line faults are generally categorized into two types: short circuit (shunt faults) and series (open conductor faults). Shunt faults occur when there is contact between two or more conductors or between a conductor and the ground. In contrast, open conductor faults occur due to a break in the conductor. These faults can arise from various causes such as lightning strikes, short circuits between lines, accidents, unforeseen incidents, or human error³.

The advancements in Artificial Intelligence (AI) have brought about significant improvements in the detection and classification of faults in transmission lines. AI techniques have demonstrated their capability to enhance the accuracy and speed of fault detection and classification, which is essential for maintaining the stability and reliability of power systems^{4–6}. AI-driven methods leverage advanced algorithms and machine learning for automatic fault detection and classification, offering improved accuracy and efficiency over traditional approaches. These methods incorporate optimization techniques for fine-tuning, enhancing both training effectiveness and overall system performance⁷. AI models identify real-time anomalies, enabling swift

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responses to critical events, pre-emptive maintenance, and risk mitigation. Furthermore, AI streamline's fault classification, reducing reliance on manual analysis and expertise^{8,9}. By analyzing patterns in sensor readings, performance metrics, and historical data, AI algorithms accurately classify faults such as line-to-ground, line-to-line, and open conductor faults.

Integrating AI into fault identification and classification offers numerous benefits for the transmission line industry. Automating these processes significantly reduces human error and response time, enhances system reliability, and minimizes downtime. AI models continuously adapt to evolving conditions, improving fault detection accuracy over time^{10,11}. As data availability increases and AI technologies advance, AI-driven fault identification and classification will play an essential role in ensuring the seamless operation of electrical grids, driving efficiency, reliability, and innovation in the industry¹². AI can greatly enhance fault detection in VSC-based MTdc systems by improving adaptability and reliability in protection strategies, addressing grid complexities, and supporting renewable energy integration¹³. A model for locating dc-link faults in MT-HVdc networks combines DWT, Bayesian optimization, and ANN, achieving robust detection even in noisy environments¹⁴. ANFIS effectively integrates ANN learning with fuzzy logic for fault location and classification^{3,15,16}. A deep learning-based protection scheme for meshed high-voltage direct current (HVDC) grids employs LSTM networks and discrete wavelet transforms to enhance fault detection reliability without complex thresholding¹⁷.

In 2021,¹⁸ developed a robust disturbance classification method addressing PMU data quality issues, using a univariate temporal convolutional denoising autoencoder (UTCN-DAE) and a multivariable temporal convolutional denoising network (MTCN) for feature extraction, achieving 97.69% accuracy with high computational efficiency. Similarly,¹⁹ presented a fully automated deep-learning approach for fault classification in grid networks using a fully convolutional network (FCN). Tested on an IEEE 30-bus system, their method achieved an impressive accuracy of 99.27%, outperforming existing methods and validated through tenfold cross-validation and various performance metrics.

In Reference³³, Tong et al. introduced an intelligent fault diagnosis approach for rolling bearings that utilizes Gramian Angular Difference Field (GADF) and an improved dual attention residual network. This method highlights the potential of advanced feature extraction techniques, such as GADF, in capturing complex spatiotemporal patterns from sensor data, which can be beneficial for improving fault classification accuracy in power transmission line systems. The dual attention mechanism used by Tong et al. provides an additional layer of feature prioritization, which could inspire further refinement of fault detection models, particularly in identifying the most relevant features in noisy, multidimensional datasets. In Reference³⁴, Ma et al. explored relaying-assisted communications for demand response in smart grids, incorporating cost modeling, game strategies, and algorithms. While this paper primarily addresses communication systems, the underlying optimization strategies for resource allocation and decision-making in smart grids offer valuable insights for enhancing the coordination between fault detection systems and grid management tools. Such strategies could be adapted to improve the efficiency and speed of fault detection responses in transmission line systems. Reference³⁵ by Hang et al. presented a method for diagnosing interturn short-circuit faults and fault-tolerant control in DTP-PMSMs using subspace current residuals. This work emphasizes the importance of fault-tolerant control, which is highly relevant for power transmission systems where ensuring continued operation in the event of faults is critical. The use of current residuals for fault detection in rotating machinery can provide a framework for integrating fault-tolerant mechanisms with fault detection algorithms, enhancing the robustness of the overall system. Similarly, Reference³⁶ by Hang et al. proposed an improved fault diagnosis method for permanent magnet synchronous machine (PMSM) systems using lightweight multi-source information data layer fusion. This approach underlines the benefit of combining multiple data sources to improve fault diagnosis accuracy. The concept of multi-source data fusion is particularly relevant for power transmission systems, where data from various sensors and sources can be integrated to strengthen fault detection and classification models. Finally, in Reference³⁷, Ni et al. developed an explainable neural network by integrating the Jiles-Atherton model and nonlinear auto-regressive exogenous models for modeling universal hysteresis. This work contributes to the growing field of explainable AI by focusing on model transparency and interpretability, which is essential for high-stakes applications like fault detection in power systems. The integration of explainable AI in fault detection systems could enhance the trust and adoption of machine learning-based approaches by providing more transparent insights into the decision-making process of the models.

Addressing the challenge of limited labeled fault data in power distribution systems,²⁰ proposed a novel multi-task learning framework that leverages latent structures in unlabeled data, achieving an accuracy of 99.02% on distribution-level phasor devices and demonstrating robustness against measurement noise. Reference²¹ introduced a method for fault diagnosis of power transformers using dissolved gas analysis (DGA) and data transformation techniques, employing six optimized machine learning methods to achieve a high predictive accuracy of 90.61%, thereby enhancing power system reliability through early fault detection.

Reference²² further advanced the field with a 3D deep learning algorithm for accurately classifying power system faults using RGB channels from 4D images of transformed fault currents. Their model, which addressed overfitting with dropout provisions, achieved classification accuracies of 93.75% and 100% with dropout values of 0.4 and 0.5, respectively. Meanwhile,²³ developed an anomaly-based technique for fault detection in electrical power systems using One-Class SVM and PCA-based models, achieving accuracies of 79.84% and 79.28%, respectively, on the VSB Power Line Fault Detection dataset from Kaggle.

In a similar²⁴ proposed an unsupervised framework utilizing a capsule network with sparse filtering (CNSF) for fault detection and classification in transmission lines. The CNSF model demonstrated high accuracy ranging from 97 to 99%, showcasing robustness against noise and high impedance faults (HIF) across different TL topologies and fault conditions. Reference²⁵ introduced open-source software for generating synthetic power quality disturbances (PQDs) to benchmark classification techniques. Their deep-learning-based

classifiers achieved accuracies of 99.28–99.75% (without noise) and 96.28–98.13% (with noise), facilitating the development and comparison of PQD classification algorithms.

Reference²⁶ proposed an LSTM-based 'end-to-end' machine learning approach for fault detection and classification in power transmission networks, achieving a high accuracy of 99.00% across diverse fault scenarios and operational conditions on the WSCC 9-bus system. Additionally, in²⁷, a CNN transformer model was introduced for detecting and localizing faults in power lines, achieving 97.53% accuracy in fault classification and 96.14% in fault localization on the IEEE 14-bus system. In²⁸, a data-driven approach for fault location and classification in power distribution systems is presented, combining wavelet transformation with optimized Convolutional Neural Networks (CNNs) and achieving fault detection accuracy of 91.4%, branch identification accuracy of 93.77%, and fault type classification accuracy of 94.93%. Meanwhile,²⁹ proposes a method for fault detection and classification in DC transmission lines using LSTM networks and discrete wavelet transform (DWT), achieving 99.04% accuracy through a unique relay system and Bayesian optimization for hyperparameter tuning. Additionally, a hybrid CNN-LSTM model integrated with real-time RFID data enhances rotor angle stability and fault detection in microgrids, achieving a classification accuracy of 94.93%³⁰.

Lastly,³¹ developed an ensemble learning model using PMU data for transmission line fault classification, achieving an accuracy of 99.88% with Explainable AI (XAI) used to interpret predictions, validated on the IEEE 14-bus system. In response to existing research that prioritizes high accuracy while complicating models with ensemble techniques, this study proposes a single optimized ensemble method, RF-LSTM Tuned KNN, which balances performance with computational efficiency. This method achieves 99.93% accuracy in multi-label fault classification, enhancing transmission system reliability. Various machine learning models, including Random Forest (RF), K-Nearest Neighbors (KNN), and Long Short-Term Memory (LSTM), were evaluated across binary and multi-label datasets, establishing a robust benchmark for fault detection.

The integration of AI techniques in power transmission systems offers several benefits, including reduced human intervention, faster response times, enhanced predictive maintenance capabilities, and improved adaptability to changing conditions. By focusing on high-performing models, this research enhances fault detection capabilities and contributes to the development of resilient and intelligent power grids while promoting efficiency and reliability.

Recent advancements in fault detection for transmission lines and power networks using machine learning and deep learning have shown significant progress, as illustrated in Table 1. These contributions range from hybrid ensemble models that combine various algorithms to deep learning approaches leveraging neural networks. Research in this field has demonstrated promising results in enhancing the accuracy and reliability of fault detection, which is crucial for maintaining the stability of power systems, given the vulnerabilities inherent in transmission line operations. The continuous innovation in this area is essential to address the evolving challenges in power system reliability and to ensure an uninterrupted electricity supply to consumers.

Despite the progress, previous studies have often concentrated on individual algorithms or lacked comprehensive ensemble approaches. Moreover, the analysis of diverse fault scenarios, especially for binary and multi-label data, has been limited, which impacts detection accuracy. This paper aims to fill these gaps by introducing a novel ensemble technique, the RF-LSTM Voting Ensemble, which significantly enhances fault detection performance. By leveraging a comprehensive dataset and evaluating multiple machine learning algorithms, this study offers critical insights into improving grid reliability and stability.

Research topic	Dataset	Modeling technique	Performance	Researcher	Year
Ensemble learning-based transmission line fault classification with explainable AI (XAI)	Multi-label	EXAI	99.88%	31	2024
Microgrid Rotor Angle Stability via RFID Data and Deep Learning	Multi-label	CNN-LSTM	99.88%	29	2024
Bayesian-optimized LSTM-DWT approach for reliable fault detection in MMC-based HVDC systems	Multi-label	LSTM-DWT	99.04%	30	2024
Fault location and classification in power distribution systems	Binary-& Multilabel	WTO-CNN	91.4% (detection) 94.93% (classification)	28	2023
Power system network's fault classification and localization	Multi-label	XT	97.53%(classification) 96.14%(localization)	27	2023
Transmission line fault detection and classification	Multi-label	LSTM	99.00% 96.28–98.13% (without noise-with noise)	26	2021
Power quality disturbances dataset generator with reference classifiers	Multi-label	CNN BiLSTM	99.28–99.75% (without noise-with noise)	25	2021
Transmission line fault detection and classification	Unlabeled	CNSF	CNSF97–99% (noise-high impedance)	24	2021
Power system network's fault detection	Binary-label	SVM PCA	79.84–79.28%	23	2021
Power system fault classification	Multi-label	DLA	93.75–100% (dropout: 0.4–0.5)	22	2021
Power transformer fault diagnosis	Multi-label	EL	90.61%	21	2021
Power distribution system fault classification	Unlabeled	MTLS-LR	99.02%	20	2021
Power system fault classification	Multi-label	CNN	99.27%	19	2021

Table 1. Overview of the recent contributions.

Data set type	Multivariate
Number of columns	10
Numbers of rows	11,701
Numerical features	I _a , I _b , I _c , V _a , V _b , V _c ,
Categorical features	G, A, B, C,
Labels	NF, L-G, LL, LL-G, LLL, LLL-G
Memory usage	63.24 KB

Table 3. RTD-dataset overview.

Data set type	Binary	Multivariate
Number of columns	9	10
Numbers of rows	12001	7861
Numerical features	I _a , I _b , I _c , V _a , V _b , V _c , unnamed,	I _a , I _b , I _c , V _a , V _b , V _c , unnamed
Categorical features	Output (S)	G, A, B, C,
Labels	NF, FF	NF, L-G, LL, LL-G, LLL, LLL-G
Memory Usage	843.9 KB	61.3 KB

Table 2. Dataset overview (from³²).

The primary objective of this research is to develop a robust fault detection and classification system for power transmission lines. We evaluated the performance of individual machine learning models, such as Random Forest, KNN, and LSTM, on both binary and multi-label datasets. We also implemented an advanced ensemble technique, the RF-LSTM Voting Ensemble, aiming to achieve high accuracy while managing model complexity. Evaluation metrics such as accuracy, precision, recall, and F1-score were utilized to assess the models’ performance.

The RF-LSTM Voting Stacked Tune KNN showed strong results, with KNN and Random Forest emerging as top performers on binary datasets. These findings highlight the effectiveness of our approach in advancing fault detection capabilities and enhancing the resilience of power grids.

Methodology

The study used machine learning algorithms such as Random Forest (RF), K-Nearest Neighbors (KNN), and Long Short-Term Memory (LSTM) to identify and classify transmission line faults. Additionally, an RF-LSTM Stacked Tune KNN Ensemble was implemented. The framework involved data pre-processing and prediction modeling, with Particle Swarm Optimization (PSO) employed for hyperparameter tuning. Models were evaluated using confusion matrices, cross-validation, ROC curves, and learning curve analysis. We used TensorFlow for deep learning to develop and train our detection and classification models. Our experiments were conducted on a system with an Intel Core i5-4590 CPU, Nvidia GeForce GTX 750Ti GPU, and 8 GB of RAM. This setup provided adequate computational resources to train, test, and evaluate the hybrid models, allowing us to effectively validate their performance in electric fault detection and classification.

Dataset analysis

The Electrical Fault Detection and Classification dataset from Kaggle is crucial for enhancing fault detection in power transmission networks. It includes line currents and voltages recorded under various fault conditions from a system with four 11 kV generators and transformers at the midpoint of the transmission line. This dataset supports the development of algorithms for accurate fault detection and classification, improving network reliability and minimizing downtime. The dataset comprises two parts: a binary classification dataset with 12,001 rows and 9 columns for detecting fault presence, and a multi-class classification dataset with 7861 rows and 10 columns for identifying specific fault types. To ensure the final model’s robustness, a real-time simulated dataset with 11,701 rows and 10 columns is generated by applying faults at different lengths of transmission lines, as shown in Table 3, is used for training and testing. This dataset reflects practical scenarios with dynamic environmental conditions and operational variations, providing a more comprehensive evaluation framework. Tables 2 and 3 outline the details of these datasets, while Table 4 lists the six fault categories, including No-Fault, Line to Ground, Line to Line, Line to Line to Ground, Line to Line to Line, and Line to Line to Line to Ground. These datasets provide a comprehensive foundation for both detecting and classifying faults in electrical relays and transmission lines.

The dataset analysis is depicted through several figures. Figure 1 shows scatter plots of voltage versus current for three lines (a, b, c), highlighting the voltage-current relationships and line-specific behaviors. Figure 2 details classification tasks: Fig. 2a represents binary classification with 54.2% “fault” and 45.8% “no-fault” instances, while Fig. 2b illustrates a multi-class classification with six fault types and a “no-fault” category, including specific distributions for each fault type. Finally, Fig. 3 displays unimodal distributions for current and voltage in both classification scenarios, showing typical values and aiding in outlier detection.

Target label	Description
NF	Balanced signal flow, no direct contact between conductors and ground
L-G	L-G A single line fault occurs when one conductor on a transmission line contacts the neutral conductor or drops to the ground
LL	LL Double line fault refers to a situation in which two conductors are short circuited
LL-G	LL-G A double line to ground fault occurs from a short circuit between two phase conductors and ground
LLL	LLL A three phase short circuit fault defines a short circuit between three phase conductors
LLL-G	LLL-G A short circuit between three phase conductors and the ground is referred to as a three phase to ground fault

Table 4. Target label information.

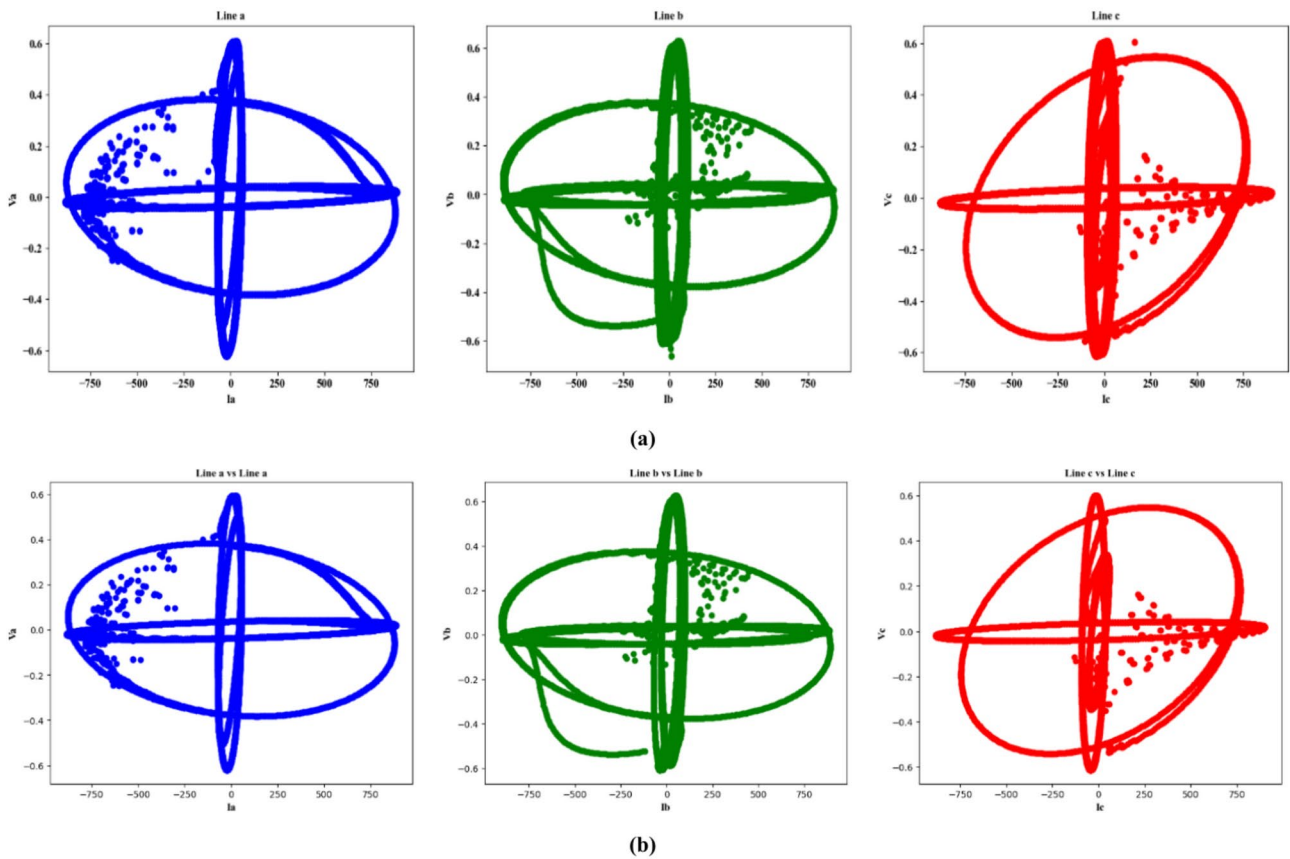


Fig. 1. Scatter plots of voltage-current relationships across transmission lines (a) Binary Classification (b) multi-classification.

Figure 3 presents correlation matrices for binary, multi-label datasets and RTD-multi-label datasets labeled as (a), (b) and (c) respectively, to identify the exact set of input parameters in machine learning. The binary data correlation matrix in (a) showcases the relationships among variables in a binary classification context, while (b) and (c) illustrate correlations within a multi-classification framework. Rows and columns correspond to variables, and the values indicate correlation coefficients, reflecting the strength and direction of relationships. Diagonal elements, always displaying a correlation of 1, represent self-correlation, while off-diagonal elements reveal pairwise correlations. Values range from -1 to 1 : a correlation of 1 signifies a robust positive relationship, -1 represents a strong negative correlation, and 0 indicates no correlation. These correlation matrices uncover patterns, identify potential multi collinearity issues, and provide insights into the dataset’s structure. However, correlation does not imply causation, necessitating further analysis to establish definitive causal relationships.

Data preprocessing

After analyzing the dataset’s characteristics, it was divided into training (75%) and testing (25%) sets, ensuring representation from each scenario. The target column was label encoded to a numeric format. Principal Component Analysis (PCA) was applied to input features to reduce noise and correlation, enhancing model performance by streamlining feature space dimensionality. In binary classification, PCA highlighted features closely linked to fault occurrences, while in multi classification, six principal components captured the dataset’s

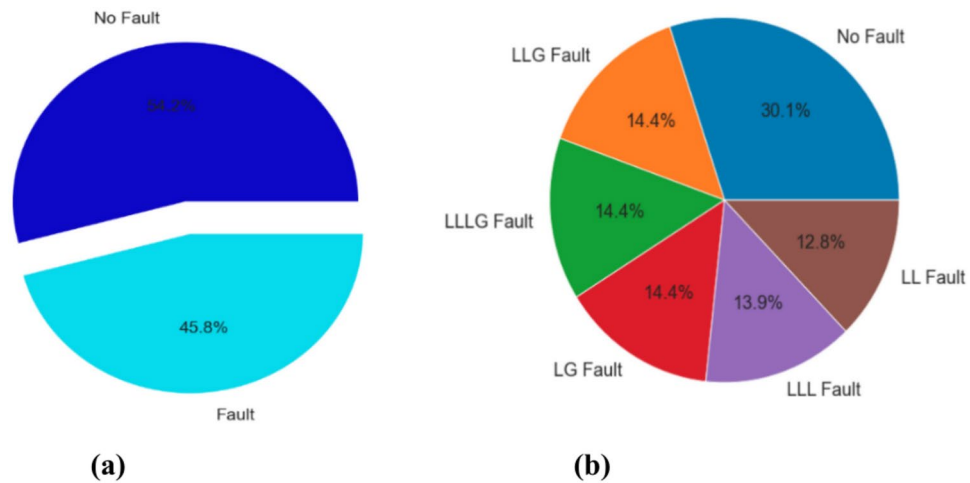


Fig. 2. Distribution of fault categories in (a) Binary and (b) Multi-Label Classification Tasks.

variance, retaining essential information for accurate predictions. Representation of the data preparation stages for input features is depicted in Fig. 4.

To address the class imbalance, the Synthetic Minority Over-Sampling Technique (SMOTE) generated synthetic samples for minority class instances using Eq. 1.³¹

$$X_g = X_m + \delta(X_n - X_m) \quad (1)$$

where δ is a random number in the range $[0, 1]$. This balanced the training data and improved model accuracy. Feature standardization was achieved using z-score normalization (Eq. 2):

$$z = \frac{X - \mu}{\sigma} \quad (2)$$

ensuring features had a mean of zero and a standard deviation of one, aiding in model training. Supervised classifiers were trained and evaluated on binary and multi-class datasets. To mitigate overfitting due to the small dataset size, fourfold cross-validation and hyperparameter tuning using the Particle Swarm Optimization (PSO) algorithm (Eqs. 3 and 4) were employed.

$$v_{(i,j)}^{(t+1)} = w \cdot v_{(i,j)}^t + c_1 r_1 (p_{(i,j)} - v_{(i,j)}^t) + c_2 r_2 (g_j - x_{(i,j)}^t) \quad (3)$$

$$x_{i,j}^{t+1} = x_{i,j}^t + v_{i,j}^{t+1} \quad (4)$$

In PSO for hyperparameter tuning, each particle's position x represents a set of hyperparameters. The algorithm iteratively updates each particle's position and velocity, guided by its personal best position p and the swarm's global best position g , to find the optimal hyperparameters for improved model performance.

Proposed framework

The proposed framework first detects faults and then classifies them into different categories. This section outlines the methodology for binary classification, which was then refined for multi-class classification, as shown in Fig. 5. At this stage, the dataset has already been pre-processed and split into training and testing sets.

The initial datasets, D_{binary} and D_{multi} containing binary and multi-label classifications respectively, are described in Eq. 5. To ensure independent and identically distributed data for cross-validation and evaluation, each dataset was divided into four subsets, denoted as follows:

$$D_{binary_i} = \{x_i, y_i\} \text{ for } i = 1, 2, 3, 4, \dots \quad (5a)$$

$$D_{multiclass_i} = \{x_i, y_i\} \text{ for } i = 1, 2, 3, 4, \dots \quad (5b)$$

In this iterative process, the dataset is partitioned into four subsets over four rounds ($j = 1, 2, 3, 4$). In each round, three subsets form the training set D_{train}^j , while the remaining subset acts as the test set, as shown in Eq. 6.

$$D_{train}^j = D_{binary_i} \cup D_{binary_{i+1}} \cup D_{binary_{i+3}} \{j = 1, 2, 3, \dots, i, i+1, i+2 \neq j\} \quad (6a)$$

$$D_{test}^j = D_{binary_j} \{j = 1, 2, 3, \dots\} \quad (6b)$$

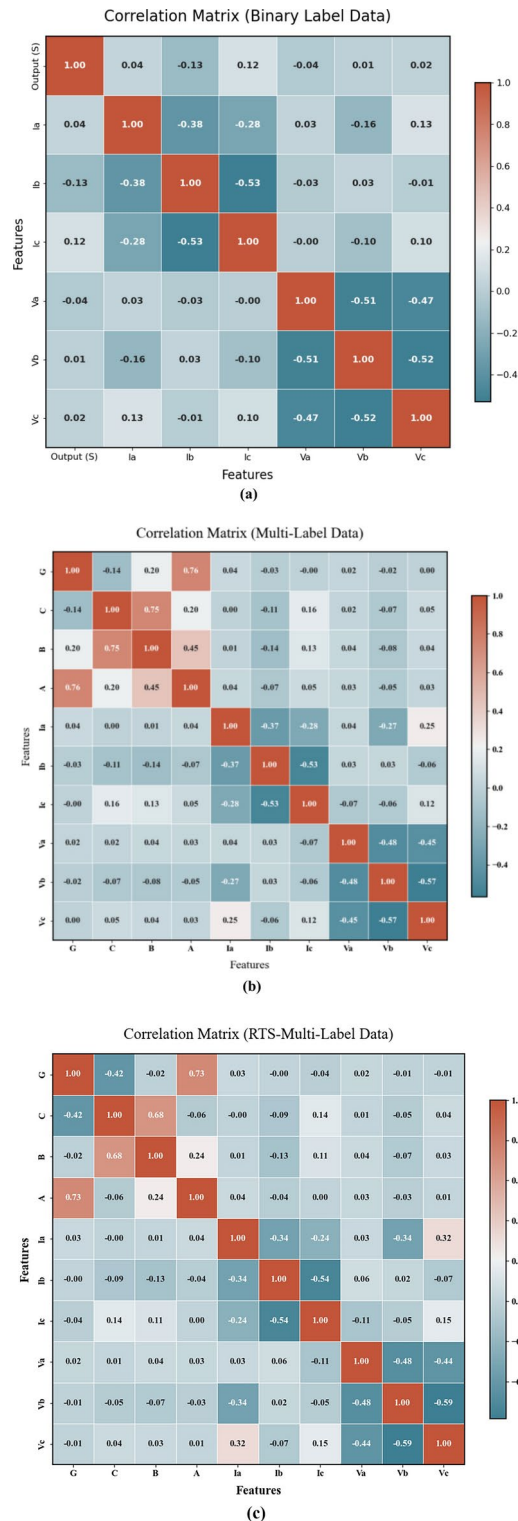


Fig. 3. Correlation matrices (a) Binary Label dataset (b) Multi-Label Dataset (c) RTS-Multi-Label Dataset.

The same partitioning process applies to the multi-label classification dataset D_{multi} , ensuring each subset is used once as the test set for robust cross-validation. We applied three classification models to both datasets: Random Forest (RF), K-Nearest Neighbors (KNN), and Long Short-Term Memory (LSTM).

Random Forest constructs multiple decision trees. For binary classification, the final prediction $y_{RF(x)}$ is determined by majority voting among M trees (Eq. 7a). For multi-label classification, the prediction $y_{RF(x)}^k$ for each label y^k is also based on majority voting among M trees (Eq. 7b).

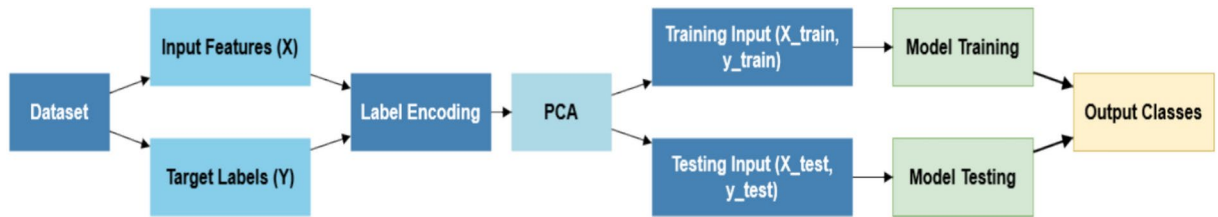


Fig. 4. Representation of the data preparation stages for input features.

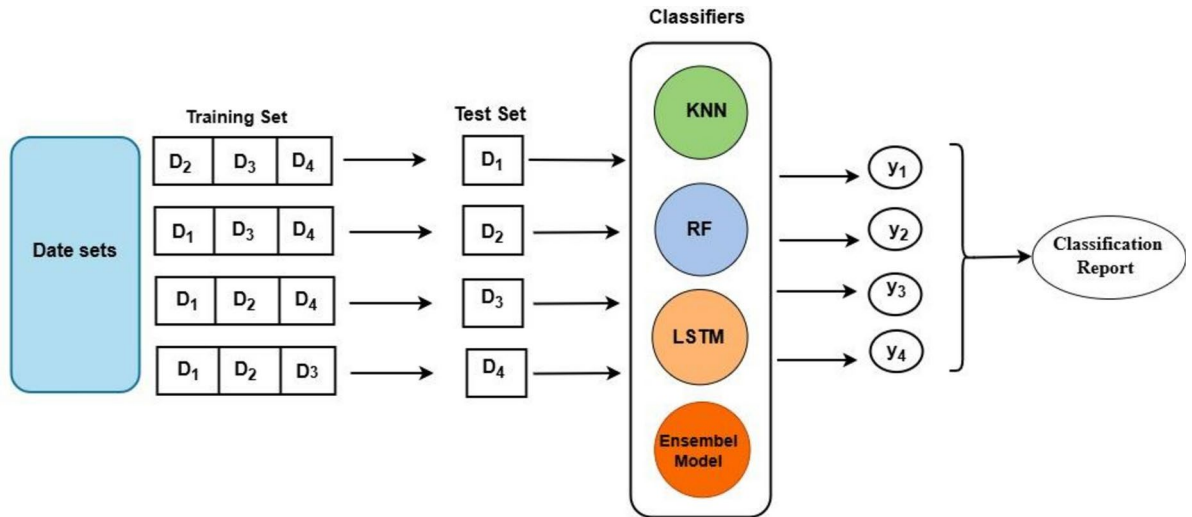


Fig. 5. Overall workflow diagram of the proposed method.

$$y_{RF}(x) = \operatorname{argmax}_{c \in \{0,1\}} \sum_{i=1}^M I(y_{RF_i}(x) = c) \quad (7a)$$

$$y_{RF}^k(x) = \operatorname{argmax}_{c \in \{0,1\}} \sum_{i=1}^M I(y_{RF_i}^k(x) = c) \quad (7b)$$

Long Short-Term Memory (LSTM) networks build a recurrent neural network during training. For binary classification, the final prediction $y_{LSTM}(x)$ is obtained by applying a sigmoid activation function σ to the output of the last time step. Specifically, this prediction is calculated using the hidden state h_t the weight matrix W_o and the bias term b_o for the output layer (Eq. 8a). For multi-label classification, the prediction $y_{GRU}(x)^k$ for each label k is similarly determined by applying σ to the output of the last time step for that label, using the hidden state h_t weight matrix W_o^k and bias term b_o^k (Eq. 8b).

$$y_{GRU}(x) = \sigma(W_o h_t + b_o) \quad (8a)$$

$$y_{GRU}(x)^k = \sigma(W_o^k h_t + b_o^k) \quad (8b)$$

k-Nearest Neighbors (k-NN) predicts based on the labels of the k closest data points. For binary classification, the final prediction $y_{KNN}(x)$ is determined by majority voting among the k nearest neighbors, where $y_{KNN_i}(x)$ is the label of the i -th nearest neighbor and I is the indicator function (Eq. 9a). For multi-label classification, the prediction $y_{KNN}(x)^k$ for each y^k is also determined by majority voting among the k nearest neighbors for that label (Eq. 9b).

$$y_{KNN}(x) = \operatorname{argmax}_{c \in \{0,1\}} \left(\sum_{i=1}^k I(y_{KNN_i}(x) = c) \right) \quad (9a)$$

$$y_{KNN}(x)^k = \operatorname{argmax}_{c \in \{0,1\}} \left(\sum_{i=1}^k I(y_{KNN_i}^k(x) = c) \right) \quad (9b)$$

After the analysis of the individual models Random Forest (RF) and Long Short-Term Memory (LSTM) the strengths and limitations in multi-fault classification were identified. To enhance overall performance, a stacked tune ensemble technique is implemented, combining the predictions of both models for each label. This

ensemble approach integrates the strengths of each model, resulting in more robust and accurate classifications. In multi-fault classification, the RF-LSTM Stacked Tune KNN ensemble model combines predictions from Random Forest and LSTM. Let $y_{RF(x)}^k$ and $y_{LSTM(x)}^k$ denote the predictions for the k-th fault label. The final prediction $y_{stack}(x)^k$ is determined as in Eqs. 10 and 11:

$$z(x)^k = (y_{RF(x)}^k, y_{LSTM(x)}^k) \quad (10)$$

$$y_{stack}(x)^k = \underset{c \in (0,1)}{\operatorname{argmax}} \sum_{i=1}^k I(y_{stack}(x)^k = c) \quad (11)$$

Here, $y_{stack}(x)^k$ is the final prediction for the k-th label, I is the indicator function that equals 1 if the statement is true and 0 otherwise, and $\underset{c \in (0,1)}{\operatorname{argmax}}$ selects the class c that is predicted by the majority of the nearest neighbors.

Evaluation of the methodology

To evaluate the performance of a classification model, we used several metrics, including the confusion matrix, accuracy, precision, recall, F1 score, and ROC curve with AUC. The confusion matrix is a table that shows the counts of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN), providing a comprehensive overview of the model's performance. Accuracy is calculated as the ratio of correctly predicted instances (both TP and TN) to the total number of instances, as shown in Eq. 12^{8,32}.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (12)$$

Accuracy provides the overall correctness of the model but may be misleading in scenarios with imbalanced datasets, where certain fault types are more frequent than others. Precision measures the accuracy of positive predictions and is crucial when the importance of false positives is high, such as incorrectly identifying a fault that isn't there, as shown in Eq. 13. Recall, or sensitivity, as shown in Eq. 14, evaluates the model's ability to correctly identify actual faults. This is vital when the importance of missing a fault (false negatives) is high, as it could lead to severe consequences in the power system's operation. The F1 score balances precision and recall, providing a single metric that is useful when dealing with imbalanced classes of faults as explained in Eq. 15.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (13)$$

$$\text{Recall} = \frac{TP}{TF + FN} \quad (14)$$

$$\text{F1score} = 2 * \frac{\text{Precision} \times \text{Recall}}{\text{recision} + \text{Recall}} \quad (15)$$

$$\text{FPR} = \frac{FP}{FP + TN}, \text{TPR} = \frac{TP}{TP + FN} \quad (16)$$

$$\text{AUC} = \sum_{i=1}^{n-1} \left[(\text{FPR}_{i+1} - \text{FPR}_i) \cdot \frac{\text{TPR}_i + \text{TPR}_{i+1}}{2} \right] \quad (17)$$

The Receiver Operating Characteristic (ROC) curve plots the true positive rate (Recall) against the false positive rate (FPR), as described in Eq. 16. The Area Under the ROC Curve (AUC), defined in Eq. 17, summarizes the model's performance, with an AUC of 1 indicating a perfect model and 0.5 indicating no discriminative ability. These metrics provide a comprehensive assessment of our model's performance, ensuring robustness, reliability, and effectiveness in real-world fault identification and classification for transmission lines.

Results and discussions

Following the discussion on the model and its mathematical framework, this section detailed the evolution of our proposed methodology, beginning with binary classification and advancing to multi-class classification. It thoroughly examined the performance and results using comprehensive evaluation metrics, providing a clear context for our analysis.

Binary classification

Outstanding performance and reliability in binary fault classification have been demonstrated by the models. Detailed evaluations and comparisons underscore the strengths of each model in accurately identifying faults. As illustrated in Fig. 6, near-perfect performance is revealed by the confusion matrices. Notably, remarkable metrics were achieved by the Random Forest (RF) model, with an accuracy of 99.72%, precision of 99.72%, recall of 99.65%, and an F1 score of 99.78%. Slightly better performance was exhibited by the K-Nearest Neighbors (KNN) model, with an accuracy of 99.75%, while impressive performance was also demonstrated by the LSTM model, with an accuracy of 98.70%. These metrics underscore the robust capabilities of all models in binary fault classification.

In addition to high accuracy, precision, recall, and F1 scores, the performance of the models is further emphasized by ROC curves and comprehensive analyses. As shown in Fig. 7, an AUC of 1.0 is achieved by each

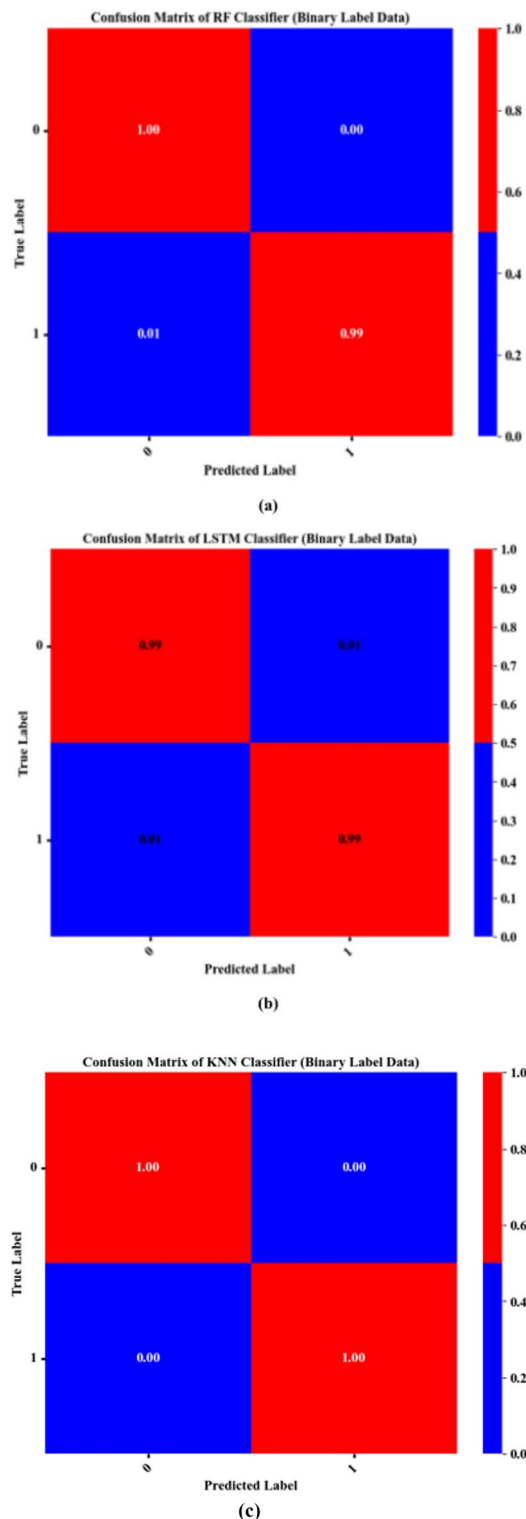


Fig. 6. The Confusion matrix of classifier models on binary label data sets: (a) RF; (b) LSTM; (c) KNN.

ROC curve, indicating perfect classification ability for the Random Forest, K-Nearest Neighbors, and LSTM models. Despite slightly lower overall accuracy, the LSTM model maintains an AUC of 1.0, showcasing strong classification capability.

A detailed analysis of each model's effectiveness in binary fault classification is provided in Table 5, revealing minimal errors and excellent discrimination between fault and no-fault cases. These findings collectively highlight the exceptional performance, high effectiveness, and reliability of the models in binary fault detection.

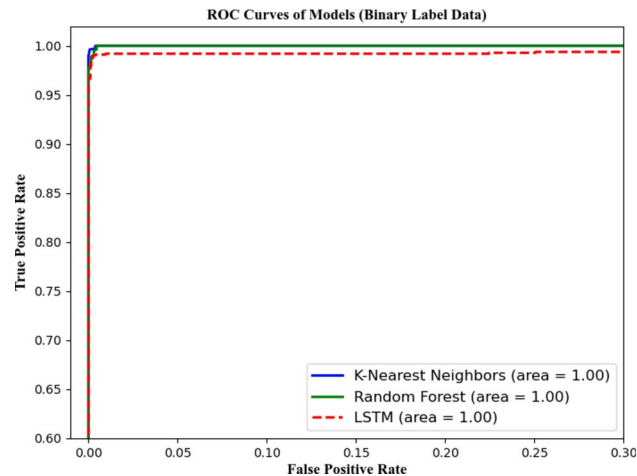


Fig. 7. The ROC curve of classifier models on binary label data sets.

Classifiers	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	Mean CV (%)
RF	99.72	99.72	99.65	99.78	99.60
KNN	99.85	99.84	99.63	99.68	99.86
LSTM	98.70	98.37	98.82	98.84	99.53

Table 5. Performance of classifier on binary label data set.

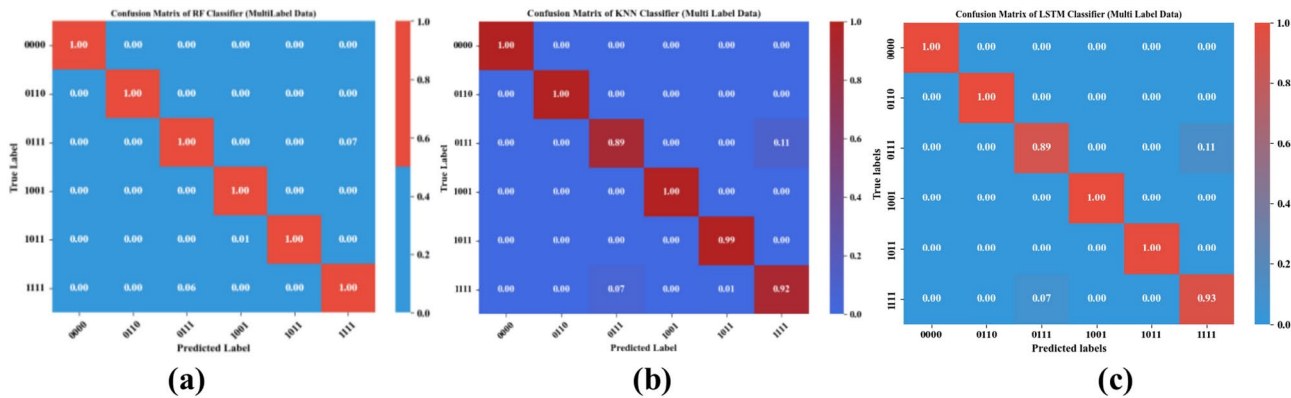


Fig. 8. The Confusion matrix of classifier models on multi label data sets: (a) RF; (b) KNN; (c) LSTM.

Multi-label classification

After the models were evaluated for binary classification, the analysis was extended to multi-class scenarios to assess their performance in more complex tasks involving multiple fault categories. An RF-LSTM Voting Ensemble model is also proposed to further enhance fault detection accuracy and robustness. The confusion matrices in Fig. 8 align closely with the ROC values in Fig. 9, providing detailed insights into each model's performance. Notably, the Random Forest (RF) and LSTM models consistently achieved impressive ROC values of 0.99 for classes 2 and 5, and perfect ROC values of 1 for other classes. High accuracy was demonstrated, with RF achieving 97.50% and LSTM achieving 96.62%. Precision, recall, and F1 scores were also notable, reflecting the models' ability to maintain high performance across different fault categories.

In contrast, slightly lower ROC values of 0.94 and 0.95 were observed for classes 2 and 5, respectively, when using the K-Nearest Neighbors (KNN) model. Despite this, the KNN model still achieved respectable accuracy (96.55%) and other performance metrics, showcasing its capability in fault classification. A comprehensive summary of the models' performance metrics, including accuracy, precision, recall, F1 score, and mean cross-validation (CV) score, is provided in Table 6.

The performance of the ensemble model, as illustrated by the confusion matrix in Fig. 10 and the ROC curve in Fig. 11, demonstrates its exceptional capabilities. A comprehensive evaluation, detailed in Table 6, reveals that this RF-LSTM Stacked Tune KNN ensemble model surpasses the individual Random Forest (RF) and Long

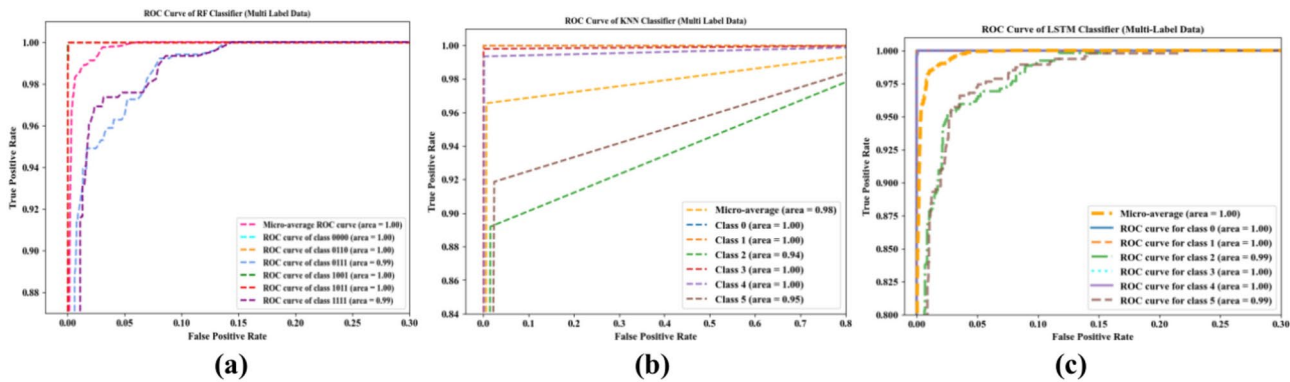


Fig. 9. The ROC curve of classifier models on multi-label data sets: (a) RF; (b) KNN; (c) LSTM.

Classifiers	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	Avg. CV (%)
RF	97.50	97.50	97.50	97.50	97.15
KNN	96.55	96.57	99.63	96.87	96.34
LSTM	96.62	96.65	96.62	96.62	95.24
RF-LSTM Stacked Tune KNN	99.96	99.96	99.93	99.93	99.96

Table 6. Performance of classifier on multi-label data.

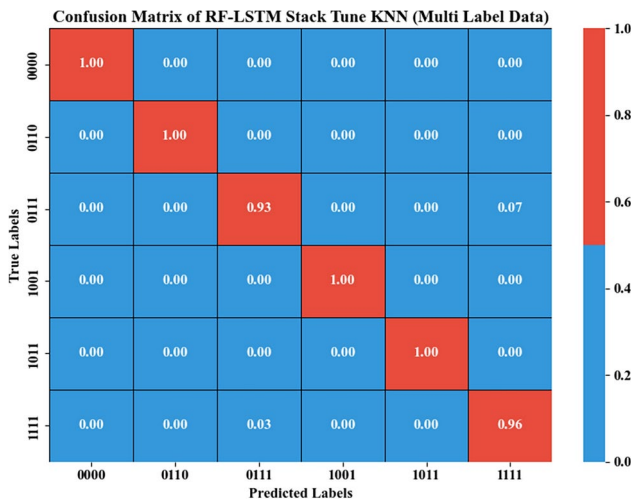


Fig. 10. The Confusion matrix for RF- LSTM Stacked Tune KNN ensemble models on multi-label data set.

Short-Term Memory (LSTM) models. While the RF and LSTM models perform well individually, the RF-LSTM Stacked Tuned KNN ensemble achieves an impressive accuracy of 99.96% and a mean cross-validation score of 98.96%, which closely aligns with the model's performance on Kaggle standard data. This result underscores the effectiveness of ensemble learning, where the combination of diverse models leverages their collective strengths, leading to robust and reliable predictions for transmission line fault classification.

The learning curves of the K-Nearest Neighbors (KNN) model and the RF-LSTM Stacked Tuned KNN model are demonstrated to showcase superior performance across different data types. The KNN model is shown to excel with binary label data, while the RF-LSTM Stacked Tuned KNN model exhibits outstanding performance with multi-label data. Both models maintain consistent accuracy and efficient learning progression over multiple epochs and data splits, validating their robustness and effectiveness.

In Fig. 12, the normalized learning curve of the KNN model is illustrated. The X-axis represents the size of the training set, which ranges from 0 to 1, while the Y-axis displays accuracy, ranging from 0.7 to 1, for improved visualization.

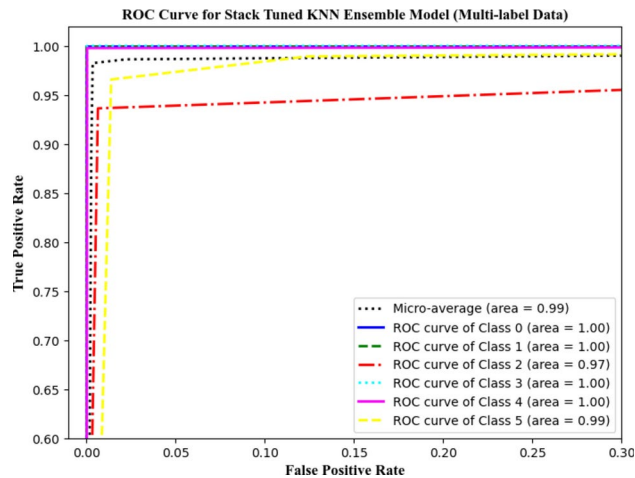


Fig. 11. The ROC Curve of RF-LSTM Stacked Tune KNN ensemble models on multi-label data set.

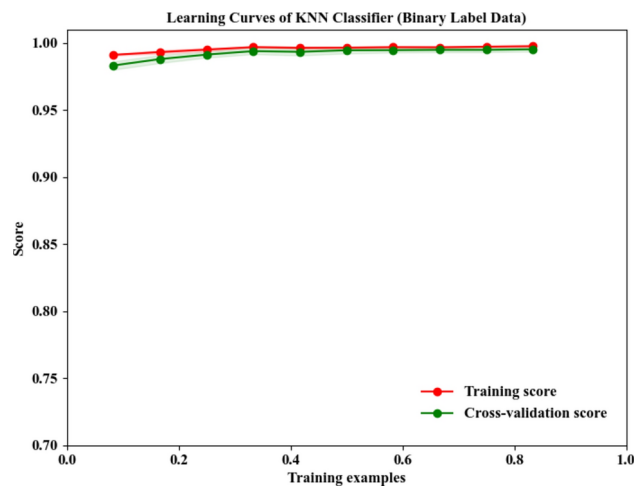


Fig. 12. Learning Curve of KNN models on Binary label data set.

Figure 13 depicts the normalized learning curve of the RF-LSTM Stacked Tuned KNN model, where the X-axis also represents the size of the training set, ranging from 0 to 1, and the Y-axis shows accuracy, ranging from 0.9 to 1, to enhance visualization.

To evaluate the robustness of the RF-LSTM Stacked Tuned KNN ensemble model for practical applications, the model is trained on real-time simulation data (RTD), as presented in Table 3. On this dataset, the ensemble achieves a remarkable accuracy of 99.97% and a cross-validation score of 99.96%, with a training time of 14 min, as shown in Table 6. The performance on RTD data is illustrated in the confusion matrix and the ROC curve shown in Fig. 14, highlighting the model's exceptional capability in accurately classifying transmission line faults. An in-depth assessment provided in Table 6 confirms that the RF-LSTM Stacked Tuned KNN ensemble model maintains high accuracy on both the Kaggle dataset and the real-time simulation dataset, demonstrating robustness and reliability across various fault categories. This affirms the model's practical applicability in real-world fault detection scenarios.

Table 7 presents the memory usage and training time for various models in both multi-label and binary classification tasks. The Random Forest (RF) model demonstrates efficiency with a training time of 17 s and memory usage of 13 MB for binary classification, while the K-Nearest Neighbors (KNN) model takes 2.3 min and uses 197 MB. The LSTM model, although powerful, requires 7.3 min and 983 MB of memory. In the multi-label scenario, RF requires 17 s and 171 MB, while KNN takes 3 min and consumes 291 MB. The LSTM model's training time increases to 9.4 min with memory usage of 731 MB. In contrast, the RF-LSTM Stacked Tuned KNN ensemble achieves a training time of 14 min and memory usage of 1.3 GB. This ensemble effectively balances accuracy and resource utilization, making it suitable for complex classification tasks, thereby highlighting its efficiency and scalability in real-world applications.

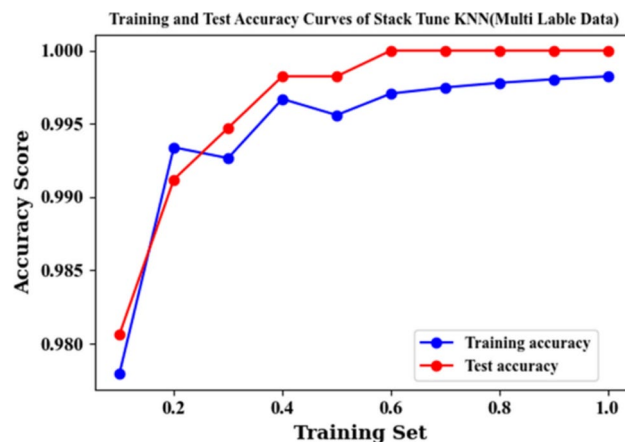


Fig. 13. Learning Curve of RF-LSTM Stacked Tune KNN ensemble models on multi-label data set.

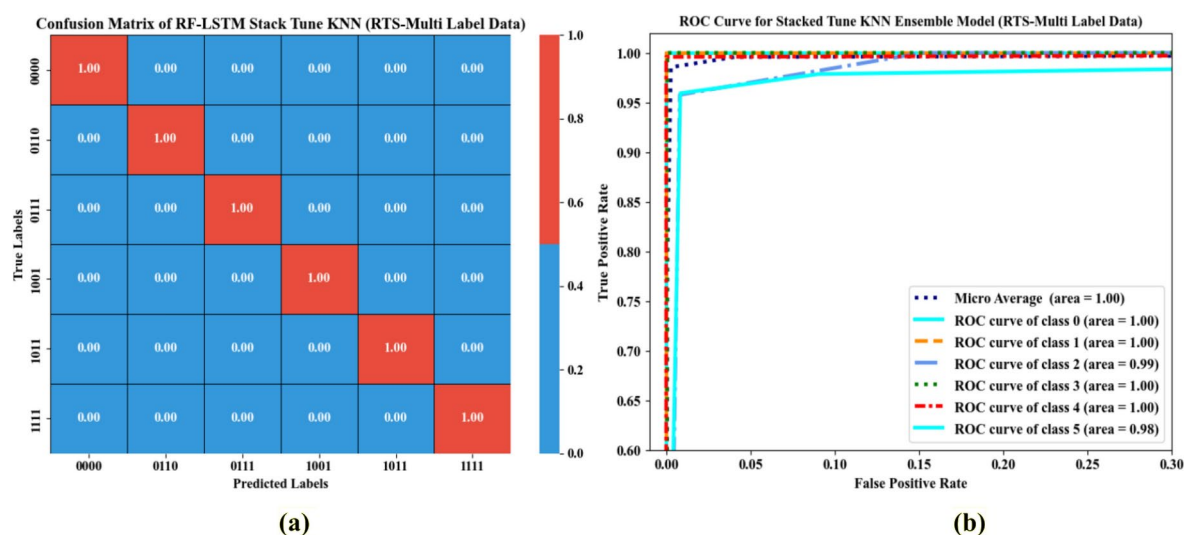


Fig. 14. Performance of RF-LSTM Stacked Tune KNN ensemble models on RTS- multi-label data set: (a) Confusion Matrix; (b) ROC Curve.

Model	Multi label data set		Binary label data set	
	Training time	Memory used	Training time	Memory used
RF	17 Seconds	171 MB	13 Second	123 MB
KNN	3 Minutes	291 MB	2.3 Minutes	197 MB
LSTM	9.4 Minutes	731 MB	7.3 Minutes	983 MB
RF- LSTM Stacked Tune KNN	14 Minutes	1.3 GB		

Table 7. Comparison of models' training time and memory usage.

Discussion

This study investigates the effectiveness of various machine learning techniques for fault classification in transmission lines, emphasizing both binary and multi-label scenarios. Our systematic evaluation focuses on three distinct models—RF (Random Forest), LSTM (Long Short-Term Memory), and KNN (K-Nearest Neighbors)—across these classification tasks. Each model's performance and suitability for detecting faults in power system data are thoroughly tested.

In binary classification, our KNN model achieves an impressive accuracy of 99.85%, significantly outperforming traditional techniques like SVM PCA, which reported an accuracy of 79.84%. This highlights the efficacy of simpler models in achieving high accuracy with reduced complexity.

Research topic	Dataset	Modeling technique	Performance	Researcher	Year
Proposed fault classification method	Multi-label	RF-LSTM Stacked Tune KNN	99.96%	Current work	2024
Proposed fault classification method	Binary-label	KNN	99.85%	Current work	2024
Transmission line faults classification with XAI	Multi-label	EXAI	99.88%	31	2024
Fault location and classification in power distribution systems	Binary and Multi-label	WTO-CNN	91.4% (Detection), 94.93% (Classification)	28	2023
Power system network's fault classification and localization	Multi-label	XT	97.53% (classification) 96.14% (localization)	27	2023
Power system network's fault detection	Binary-label	SVM PCA	79.84%	23	2021
Power system fault classification	Multi-label	CNN	99.27%	19	2021

Table 8. Comparative analysis of methodologies: literature review vs our fault classification models.

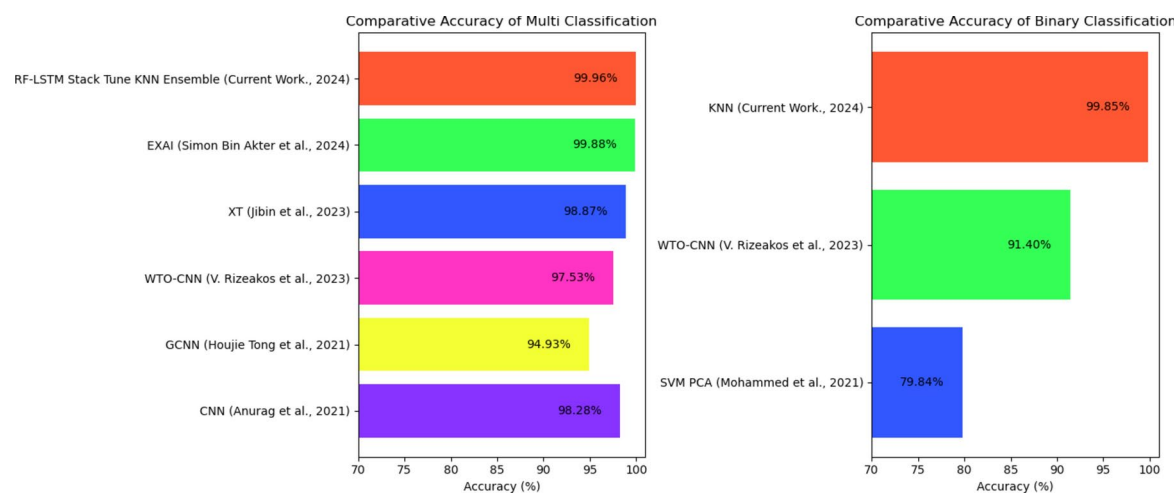


Fig. 15. Visualization of fault classification methodologies: Literature vs. Current study.

For multi-label classification, we utilize advanced ensemble techniques to enhance predictive performance. Specifically, we combine RF and LSTM models using the RF-LSTM Stack Tune KNN method. This approach excels with an outstanding accuracy of 99.96%, demonstrating its capability to handle complex datasets and achieve superior performance compared to other methodologies.

To provide deeper insights into model performance, our evaluation includes ROC curves and confusion matrices. The ROC curves (see Figs. 7, 9, 11) graphically illustrate the trade-off between the true positive rate and false positive rate, with higher Area Under the Curve (AUC) values indicating better discriminatory performance. Our RF-LSTM Stack Tune KNN model exhibits strong AUC values, further validating its effectiveness in accurately distinguishing fault classes. Additionally, the confusion matrices (see Figs. 6, 8, 10) offer a detailed view of model performance across different fault categories, showcasing the robustness of our approach.

Table 8 presents a comparative analysis of fault classification models applied to power systems, emphasizing the performance of the models alongside recent studies. In multi-label classification, the RF-LSTM Stacked Tune KNN ensemble model is shown to significantly outperform traditional deep learning approaches such as CNN (99.27%)¹⁹ and WTO-CNN (94.93%)²⁸, as well as other ensemble methods like EXAI (99.88%)³¹. These findings highlight the strategic advantage of combining RF and LSTM models within ensemble frameworks, which not only improve predictive accuracy but also reduce computational complexity in comparison to more complex deep learning architectures. Furthermore, Fig. 15 visually depicts the comparative performance of both multi-label and binary-label models, reinforcing the benefits identified in this analysis.

Furthermore, this study advance's fault classification methodologies by demonstrating the effectiveness of selective model integration and ensemble learning techniques tailored to the characteristics of power system data. The RF-LSTM Stacked Tuned KNN model is established as a new benchmark for fault classification accuracy, emphasizing its practical applicability in real-world scenarios. To evaluate the robustness of this ensemble model for practical applications, it is trained on real-time simulation data (RTD), achieving a remarkable accuracy of 99.96% and a cross-validation score of 99.68%, as shown by confusion matrix and ROC curve in Fig. 14. This ensemble model demonstrated a training time of 14 min and utilized 1.3 GB of memory, highlighting its efficiency. In conclusion, the strategic combination of RF and LSTM models proves instrumental in achieving superior fault classification performance across binary and multi-label tasks. Through the application of ensemble learning, accuracy is enhanced while operational efficiency is ensured, presenting a promising solution for reliable power system fault detection and classification.

Conclusion

This study has demonstrated the transformative impact of machine learning on fault detection in transmission lines, which is essential for enhancing the reliability and efficiency of power grids. The performance of individual models, such as Random Forest (RF), Long Short-Term Memory (LSTM), and K-Nearest Neighbors (KNN), as well as innovative ensemble techniques, was rigorously evaluated to identify optimal strategies for both binary and multi-label fault classification tasks. In binary classification, simpler models like KNN achieved exceptional accuracy, reaching 99.75%, while RF demonstrated strong performance with an accuracy of 99.72%. In multi-label classification, the RF-LSTM Stack Tune KNN model excelled, delivering a remarkable accuracy of 99.93%. This approach demonstrated the effectiveness of combining RF and LSTM models to handle complex datasets and temporal dependencies, achieving higher performance than models discussed in prior research. The use of ROC curves and confusion matrices further validated the robustness of the models, highlighting their ability to accurately distinguish fault classes and offering deep insights into model performance. In conclusion, this research advances fault classification methodologies by demonstrating the benefits of selective model integration and ensemble learning, specifically tailored to the characteristics of power system data. The strategic combination of RF and LSTM models has proven to be a practical and effective solution for achieving superior fault classification accuracy and operational efficiency.

However, there are limitations to consider ensemble models such as RF-LSTM Stack Tune KNN require large training datasets and longer training times, which can increase computational demands. This can be a challenge in real-world applications where real-time fault detection is critical, particularly in large-scale power systems. Additionally, the complexity of ensemble methods may limit their accessibility for systems with constrained computational resources. Despite these limitations, the findings present a promising path forward for enhancing grid reliability and operational efficiency through advanced machine learning techniques.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Received: 2 November 2024; Accepted: 13 January 2025

Published online: 20 January 2025

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Author contributions

Tahir Anwar, Chaoxu Mu, Wajid Khan: Conceptualization, Methodology, Software, Visualization, Investigation, Writing—Original draft preparation. Muhammad Zain Yousaf: Data curation, Validation, Supervision, Resources, Writing—Review & Editing. Saqib Khalid, Ahmad O. Hourani, Ievgen Zaitsev: Project administration, Supervision, Resources, Writing—Review & Editing.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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